

# The $\Phi_{\text{MDL}}$ Field: Quantum Structure, Born Rule, and Continuum Completion of the Chiral Minkowski CA

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## Abstract

We study the  $\mathbb{Z}_7$ -symmetric Klein–Gordon field  $\Phi_{\text{MDL}}$  as the continuum counterpart of the Three-Layer Chiral Minkowski CA (CMCA, Paper P41). Four principal results follow. (1) *Poincaré invariance*:  $\Phi_{\text{MDL}}$  satisfies exact Lorentz invariance  $\omega^2 = k^2 + m^2$ , machine-certified via `poincare_invariance_of_kg` in `LorentzInvariance.lean` ( $\mathbf{A}_{\text{Lean}}$ , zero sorry). (2) *Born rule from field amplitudes*: the position-space probability density  $P(x) = |\partial_x \Phi|^2 / \int |\partial_x \Phi|^2$  normalizes exactly ( $\mathbf{A}/\mathbf{D}$ ); extended to overlapping multi-kink configurations via the canonical dual-frame inner product (`born_rule_from_dual_frame_master` in `DualFrameBornRule.lean`,  $\mathbf{A}_{\text{Lean}}$ , zero sorry). (3) *Physical quantum state*: the pre-measurement  $\Phi_{\text{MDL}}$  state is the Hamiltonian thermal ensemble  $P(k) = e^{-M_k/T}/Z_T$  restricted to PSC-admissible  $\mathbb{Z}_7$  kink sectors  $\{0, 2, 3, 4, 6\}$  (`phimdl_thermal_state_master` in `PhiMDLThermalState.lean`,  $\mathbf{A}_{\text{Lean}}$  conditional, zero custom physics axioms); the physical state is vacuum-dominant at observed cosmic temperatures and approaches the uniform PSC-admissible distribution at early-universe temperatures. (4) *3+1D extension*:  $\Phi_{\text{MDL}}$  kinks become domain walls with tension  $\sigma = 290.10 \text{ MeV GeV}^{-2} = 7450 \text{ MeV fm}^{-2}$ ;  $\mathbb{Z}_7$  superselection is preserved; the Born rule  $P(k) = |c_k|^2$  is unchanged; and SR time dilation is exact. When two domain walls intersect in  $3 + 1$  dimensions, they form a junction line with exact dimensionless tension  $\lambda = -16/49$ , corresponding to  $-1654.77 \text{ MeV fm}$  (attractive; ratio  $|\lambda/\sigma| = 2/m_\varphi$ , pure geometry); certified in `WindingCoinDecoupling.lean` ( $\mathbf{A}_{\text{Lean}}$ , zero sorry). The continuum limit  $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$  is machine-certified in `CMCAContinuumLimit.lean`: algebraic content is  $M$ -independent, the Nyquist residual  $\varepsilon_0(M) = \pi^2/(3M^2) \rightarrow 0$  as  $M \rightarrow \infty$ , and  $\Phi_{\text{MDL}}$  is the unique terminal Level-1 Generative Triple Evolution (GTE) object. Two further Lean-certified results complete the picture: `no_finite_ca_exact_lorentz_replica` proves that every finite- $M$  discrete model has strictly positive Lorentz error ( $\mathbf{A}_{\text{Lean}}$ , zero sorry), and `commutes_with_winding_iff_diagonal` establishes that any  $\mathbb{Z}_7$ -winding-conserving coin must be diagonal, structurally precluding quantum interference in the discrete setting ( $\mathbf{A}_{\text{Lean}}$ , zero sorry). The explicit  $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$  descent map is numerically confirmed at  $M = 7$ : Cook A-glider winding profile vs. BPS kink RMSD  $= 5.34\% \leq \varepsilon_0(7) = 6.71\%$ , Pearson  $r = 0.994$ .

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# 1 Introduction

The Universal Generative Principle (UGP) Physics programme, the quantitative branch of the Reflexive Reality programme [7, 16] derives Standard Model physics from a minimal discrete arithmetic substrate. The foundational continuum object is the  $\mathbb{Z}_7$ -symmetric Klein–Gordon field  $\Phi_{\text{MDL}}$  [10], which serves simultaneously as: (i) the field-theoretic continuum limit of the cellular-automaton substrate; (ii) the quantum mechanical arena in which PSC-selected observables acquire Born-rule probabilities; and (iii) the terminal object of the GTE Level-1 category [17].

Paper P41 [17] established the *Three-Layer Chiral Minkowski CA* (CMCA) as the unique minimal discrete substrate carrying all five Level-1 GTE structural features, with the  $\Phi_{\text{MDL}}$  continuum field as its terminal object. The present paper is the companion continuum analysis: we derive the quantum structure of  $\Phi_{\text{MDL}}$  in detail, prove all four of the results sketched in P41’s Born-rule and quantum-state sections, extend to  $3 + 1$  dimensions, and certify the  $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$  continuum limit in Lean 4.

## Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A}/\mathbf{D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

**A<sub>Lean</sub>**: machine-checked exact arithmetic in Lean 4, zero **sorry**.

**A<sub>MDL</sub>**: MDL-unique within a declared expression class.

**A/D**: full analytic derivation; computationally confirmed.

**A**: computationally confirmed; a physics bridge or Lean cert remains.

**B**: calibrated reproduction.

**D**: exploratory / frontier.

**Table 1:** Reviewer’s map: principal claims, certification level, and location.

| Result                        | Status     | Claim                                                                                                                                            | Where              |
|-------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| $\mathbb{Z}_7$ -KG Lagrangian | <b>A/D</b> | $\mathcal{L} = \frac{1}{2}(\partial\Phi)^2 - V(\Phi)$ ; $V = (m_\varphi^2/49)(1 - \cos 7\Phi)$ ;<br>$m_\varphi = m_\tau = 1.77686 \text{ GeV}$ . | <a href="#">§2</a> |

(continued on next page)

(Table 1 continued)

| Result                             | Status                                   | Claim                                                                                                                                                                                                                                            | Where |
|------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| BPS kink mass                      | $\mathbf{A}_{\text{Lean}}$               | $M_{\text{kink}} = (8/49) m_\tau = 290.10$ MeV. Lean: <code>mkink_from_scc</code> (P34/P35).                                                                                                                                                     | §3    |
| Kink masses equal                  | $\mathbf{A}_{\text{Lean}}$               | All $\mathbb{Z}_7$ kink masses are equal. Lean: <code>phimdl_kink_masses_equal</code> in <code>PhimDLThermalState.lean</code> .                                                                                                                  | §3    |
| PSC admissible sectors             | $\mathbf{A}_{\text{Lean}}$               | $\{0, 2, 3, 4, 6\}$ are PSC-admissible; $\{1, 5\}$ PSC-forbidden. Lean: <code>psc_admissible_sectors</code> .                                                                                                                                    | §3    |
| Lorentz invariance                 | $\mathbf{A}_{\text{Lean}}$               | KG dispersion $\omega^2 = k^2 + m^2$ is Lorentz-invariant. Lean: <code>poincare_invariance_of_kg</code> , zero sorry.                                                                                                                            | §4    |
| PSC $\rightarrow$ Lorentz          | $\mathbf{A}_{\text{Lean}}$ (1 axiom)     | PSC Presentation Invariance forces $[\mathbf{D}]$ -selected observables Lorentz-equivariant. Lean: <code>PSCPILorentzMain.lean</code> .                                                                                                          | §4    |
| $T^{\mu\nu}$ covariance            | $\mathbf{A}/\mathbf{D}$                  | Noether tensor $T^{\mu\nu}$ transforms as a rank-2 Lorentz tensor. 400 random boosts verified; $\max  \Delta T  < 2 \times 10^{-16}$ .                                                                                                           | §4    |
| Position Born density              | $\mathbf{A}/\mathbf{D}$                  | $P(x) =  \partial_x \Phi ^2 / \int  \partial_x \Phi ^2$ ; normalizes exactly to 1. $P(0) = m_\varphi/2$ ; kink mass from $\int T_{00} dx$ .                                                                                                      | §5    |
| Sector Born rule                   | $\mathbf{A}_{\text{Lean}}$               | $P(k) =  c_k ^2$ from $\mathbb{Z}_7$ kink superselection. Lean: <code>born_rule_unconditional</code> in <code>BeableWindingPartitionInstance.lean</code> , zero custom axioms.                                                                   | §5    |
| Dual-frame Born                    | $\mathbf{A}_{\text{Lean}}$               | Overlapping kinks: Born rule via positive-definite Gram inner product. Lean: <code>born_rule_from_dual_frame_master</code> , <code>DualFrameBornRule.lean</code> , zero sorry.                                                                   | §5    |
| $[\mathbf{D}]$ -collapse           | $\mathbf{A}$                             | $[\mathbf{D}]$ -adjudication implements projective collapse to kink eigenstate with $P(j) =  c_j ^2$ ; MC at 500k trials passes $\chi^2 = 2.997$ (crit 12.592).                                                                                  | §5    |
| Thermal state (PSC-admissible)     | $\mathbf{A}_{\text{Lean}}$ (conditional) | $P(k) = e^{-M_{\text{kink}}/T}/Z_T$ on $\{0, 2, 3, 4, 6\}$ ; vacuum dominant for $T \ll M_{\text{kink}}$ . Lean: <code>phimdl_thermal_state_master</code> , zero custom physics axioms.                                                          | §6    |
| Two-function picture               | $\mathbf{A}_{\text{Lean}}$ (conditional) | Born rule $\mathcal{B}$ + transputation $\mathbb{P}^\top$ compose to give $\mathbb{P}(\text{outcome} = k \mid w) =  c_k(\mathbb{P}_D^\top(w)) ^2$ . Lean: <code>two_function_picture</code> , <code>transputation_state_selector_bundle</code> . | §6    |
| Domain wall tension                | $\mathbf{A}/\mathbf{D}$                  | $\sigma = (8/49) m_\tau^3/m_\tau^2 = 290.10$ MeV GeV $^{-2}$ ; $\sigma/A_{\text{wall}}$ recovers $M_{\text{eff}}$ to $10^{-13}$ relative error.                                                                                                  | §7    |
| 3+1D $\mathbb{Z}_7$ superselection | $\mathbf{A}/\mathbf{D}$                  | Coleman–Mandelstam argument extends to planar domain walls in 3+1D. Tunneling $\sim e^{-S_E}$ with $S_E \approx 0.163$ GeV; $\mathbb{Z}_7$ charge conserved in 3+1D.                                                                             | §7    |
| 3+1D Born rule                     | $\mathbf{A}/\mathbf{D}$                  | $P(k) =  c_k ^2$ is unchanged in 3+1D; sector Born is dimension-independent.                                                                                                                                                                     | §7    |
| SR time dilation (3+1D)            | $\mathbf{A}/\mathbf{D}$                  | $M_{\text{eff}}(v) = \gamma \sigma A_{\text{wall}}$ ; exact at six boost velocities, max relative error $2 \times 10^{-16}$ .                                                                                                                    | §7    |

(continued on next page)

(Table 1 continued)

| Result               | Status                                      | Claim                                                                                                                                                                                                                                                                                                                     | Where |
|----------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Algebraic Lifting    | $\mathbf{A}_{\text{Lean}}$                  | Every PSC-admissible SM configuration algebraically realised in $f_{\text{MDL}}$ ; $\mathbb{Z}_7$ winding, 3 generations, confinement, V–A lift to $\Phi_{\text{MDL}}$ . Lean: <code>algebraic_lifting_theorem</code> , <code>LiftingTheorem.lean</code> (P28).                                                           | §8.1  |
| Algebraic Descent    | $\mathbf{A}_{\text{Lean}}$                  | Every $F_{21}$ conservation law holds at all $M \geq 1$ ; algebraic content $M$ -independent. Lean: <code>algebraic_descent_theorem</code> , <code>AlgebraicDescentTheorem.lean</code> (P28).                                                                                                                             | §8.1  |
| SR Nyquist $M$ -dep. | $\mathbf{A}_{\text{Lean}}$                  | $\varepsilon_0(M) = \pi^2/(3M^2)$ is $M$ -dependent; does not descend. Lean: <code>sr_accuracy_depends_on_M</code> , zero sorry.                                                                                                                                                                                          | §8.1  |
| Continuum limit      | $\mathbf{A}_{\text{Lean}}$<br>(conditional) | $\varepsilon_0(M) = \pi^2/(3M^2) \rightarrow 0$ as $M \rightarrow \infty$ ; algebraic content $M$ -independent; $\Phi_{\text{MDL}}$ is unique terminal Level-1 GTE object. Lean: <code>cmca_continuum_limit_is_phimdl</code> , <code>CMCAContinuumLimit.lean</code> .                                                     | §8    |
| Descent map explicit | $\mathbf{A}$                                | Cook A-glider $\mathbb{Z}_7$ winding profile vs. $\Phi_{\text{MDL}}$ BPS kink: $\text{RMSD} = 5.34\% \leq \varepsilon_0(7) = 6.71\%$ ; Pearson $r = 0.994$ ; $Q = 1/7$ exactly. Numerical confirmation of $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$ .                                                                   | §8.2  |
| Junction tension     | $\mathbf{A}_{\text{Lean}}$                  | Two intersecting domain walls in 3+1D form a junction line; $\lambda_{\text{dim}} = -16/49$ (exact); $\lambda = -1654.77 \text{ MeV fm}$ ; $ \lambda/\sigma  = 2/m_\varphi$ (pure geometry, no free parameters). Lean: <code>phimdl_domain_wall_junction_tension_exact</code> , <code>WindingCoinDecoupling.lean</code> . | §7.6  |
| Junction attractive  | $\mathbf{A}_{\text{Lean}}$                  | $\lambda < 0$ : domain walls lower energy at intersections. Lean: <code>phimdl_junction_is_attractive</code> , zero sorry.                                                                                                                                                                                                | §7.6  |
| No-finite-CA replica | $\mathbf{A}_{\text{Lean}}$                  | For all $M > 0$ : $\varepsilon_0(M) > 0$ . Only $M \rightarrow \infty$ achieves exact Lorentz invariance; $\Phi_{\text{MDL}}$ is the unique zero-error model. Lean: <code>no_finite_ca_exact_lorentz_replica</code> , <code>CMCAContinuumLimit.lean</code> .                                                              | §9.8  |
| QCA impossibility    | $\mathbf{A}_{\text{Lean}}$                  | A $\mathbb{Z}_7$ -winding-conserving coin must be diagonal; diagonal coins decouple sectors and preclude quantum interference. Lean: <code>commutes_with_winding_iff_diagonal</code> , <code>WindingCoinDecoupling.lean</code> .                                                                                          | §9.9  |
| H3 holographic       | $\mathbf{A}$                                | Domain-wall entropy $S_{\text{wall}} \propto \sigma \times \text{Area}$ ; $\mathbb{Z}_7$ charge holographically encoded on wall boundary; Lifting Theorem realizes the bulk–boundary correspondence.                                                                                                                      | §9.7  |

## Central result

The  $\mathbb{Z}_7$ -KG field  $\Phi_{\text{MDL}}$  is the continuum completion of the Chiral Minkowski CA: it carries exact Poincaré invariance ( $\mathbf{A}_{\text{Lean}}$ ), a field-theoretically derivable Born rule ( $\mathbf{A}/\mathbf{D}/\mathbf{A}_{\text{Lean}}$ ), and the Hamiltonian thermal state of the PSC-admissible kink sector ( $\mathbf{A}_{\text{Lean}}$  conditional). In 3+1D, the 1+1D kinks become planar domain walls with tension  $\sigma = 290.10 \text{ MeV GeV}^{-2}$  and exact SR time dilation. The  $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$  continuum limit is machine-certified in Lean 4 with zero sorry.

## What this paper does and does not claim

**Established in this paper:** The  $\mathbb{Z}_7$ -KG field  $\Phi_{\text{MDL}}$  carries exact Poincaré invariance (`poincare_invariance_of_kg` in `LorentzInvariance.lean`,  $\mathbf{A}_{\text{Lean}}$ , zero sorry) and a field-theoretically derivable Born rule ( $P(x) = |\partial_x \Phi|^2 / \int |\partial_x \Phi|^2$ ,  $\mathbf{A}/\mathbf{D}$ ; extended to overlapping kinks via the dual-frame inner product, `DualFrameBornRule.lean`,  $\mathbf{A}_{\text{Lean}}$ ). The pre-measurement state is the Hamiltonian thermal ensemble on PSC-admissible sectors  $\{0, 2, 3, 4, 6\}$  (`phimdl_thermal_state_master`,  $\mathbf{A}_{\text{Lean}}$  conditional, zero custom physics axioms). In 3+1D, kinks become planar domain walls with tension  $\sigma = 290.10 \text{ MeV GeV}^{-2}$  and exact SR time dilation ( $\mathbf{A}/\mathbf{D}$ ). The  $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$  continuum limit is machine-certified (`CMCAContinuumLimit.lean`,  $\mathbf{A}_{\text{Lean}}$  conditional). The Pöschl-Teller fluctuation spectrum is  $s = 1$  reflectionless ( $\mathbf{A}_{\text{Lean}}$ , `phimdl_fluctuation_is_poschl_teller`), with quantum kink mass  $M^Q = 321.32 \pm 15.6 \text{ MeV}$  from MS-bar dimensional regularization ( $\mathbf{A}$ ); the prior log-UV cutoff value ( $M^Q = 230.43 \text{ MeV}$ ) is superseded. The Coleman–Weinberg mechanism is ruled out: the one-loop effective potential preserves the  $\mathbb{Z}_7$  vacuum degeneracy ( $\mathbf{A}$ ).

**Not claimed:** This paper does not establish the *unconditional* physical Born state: which specific  $D \in [\mathbf{D}]$  governs collapse on non-trivial records depends on Conjecture C2 (C2-CLOSURE), which remains open (§9.6). The existence of genuine 3D discrete gliders (non-planar configurations) in the three-layer CMCA was an open problem; it is resolved negatively (P43 [6]): no outer-totalistic  $\mathbb{Z}_7$  CA in 3+1D supports Class-4 behaviour. The 3+1D analysis here covers only planar domain walls. The absolute value of the kink mass  $M_{\text{kink}} = (8/49)m_\tau$  depends on the Self-Consistency Condition (SCC) identification  $m_\varphi = m_\tau$  (P34/P35,  $\mathbf{A}/\mathbf{D}$  bridge); unconditional  $\mathbf{A}_{\text{Lean}}$  for this identification is not claimed here.

**UGP Physics programme context.** This paper is part of the UGP Physics programme [16]. Companion references: P28 [7] (computational universality and Rule 110; Algebraic Lifting and Descent theorems that certify content-preservation morphisms in the GTE category of Level-1 constructions), P34 [10] (GTE-Möbius architecture, D1–D5 axioms,  $\Phi_{\text{MDL}}$  definition,  $[\mathbf{D}]$  structure), P35 [12] (GTE unification; EW scale in  $\Phi_{\text{MDL}}$  context; SCC mass identification  $m_\varphi = m_\tau$ ), P36 [9] (AFCA  $\tau_c$  inner clock, dynamics-level SR time dilation; emergent gravity on  $\Phi_{\text{MDL}}$ ), P37 [15] (Fock space over kink modes; Born rule and Fock quantization; 't Hooft cogwheel construction in  $\Phi_{\text{MDL}}$ ), P40 [5] (GF(7) polynomial universality certification used throughout), P41 [17] (Three-Layer Chiral Minkowski CA; the discrete UV completion of  $\Phi_{\text{MDL}}$ ; Section references use the notation “P41 §N”). The GTE-Möbius architecture axioms (D1–D5) established in P34 [10] are the foundational structure from which all results in this paper are derived.

**GTE-Möbius architecture terminology.** The **GTE-Möbius architecture** is a triple  $(A, e, [\mathbf{D}])$  defined in [10]:  $A$  is the PSC-admissible arithmetic layer (at Level 1, the  $\mathbb{Z}_7$  CMCA; at Level 2, the kink sector of  $\Phi_{\text{MDL}}$ );  $e$  is the evolutionary dynamics (at Level 1, the update rule  $f_{\text{MDL}}$ ; at Level 2, Klein–Gordon evolution); and  $[\mathbf{D}]$  is the *coherence measure*—a real-valued functional on candidate/record pairs that the Perfect Self-Containment (PSC) criterion uses to adjudicate which trajectories correspond to physical observations [10]. In the  $\Phi_{\text{MDL}}$  setting,  $[\mathbf{D}]$  *operates on*

kink-configuration superpositions and selects observables via the PSC superselection  $\{0, 2, 3, 4, 6\}$  and the Born rule;  $[\mathbf{D}]$  is not a term in the Klein–Gordon Lagrangian but an axiomatic overlay from the GTE–Möbius framework. A  $[\mathbf{D}]$ -selected observable is one assigned positive weight by  $[\mathbf{D}]$ . The PSC axiom D2 (Presentation Invariance) states that  $[\mathbf{D}]$  is invariant under any transformation preserving the PSC structure—from which Lorentz equivariance is derived in §4 [10]. Throughout, *machine-certification levels* follow the GTE convention:  $\mathbf{A}_{\text{Lean}}$  (Lean 4 proof, zero sorry),  $\mathbf{A}/\mathbf{D}$  (full analytic derivation; computationally confirmed),  $\mathbf{A}$  (computationally confirmed; no Lean certification yet).

## 2 The $\Phi_{\text{MDL}}$ Field

### 2.1 Lagrangian and potential

The  $\Phi_{\text{MDL}}$  field is a real scalar field  $\Phi : \mathbb{R}^{1,1} \rightarrow \mathbb{R}$  (1+1 dimensions; extended to 3+1D in §7) governed by the  $\mathbb{Z}_7$ -symmetric Lagrangian density [10, 12]:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \Phi)^2 - V(\Phi), \quad V(\Phi) = \frac{m_\varphi^2}{49}(1 - \cos(7\Phi)). \quad (1)$$

The mass parameter is set by the Self-Consistency Condition (SCC):  $m_\varphi = m_\tau = 1776.86 \text{ MeV}$ , the tau-lepton mass. This identification is derived in P34–P35 [10, 12] via the correspondence between the GTE orbit structure and Standard Model fermion masses.

The potential  $V(\Phi)$  has  $\mathbb{Z}_7$  discrete symmetry:  $V(\Phi + 2\pi k/7) = V(\Phi)$  for  $k \in \mathbb{Z}_7$ . The seven degenerate minima are located at

$$\Phi_k = \frac{2\pi k}{7}, \quad k \in \{0, 1, \dots, 6\}, \quad (2)$$

with  $V(\Phi_k) = 0$  and  $V''(\Phi_k) = m_\varphi^2$ .

### 2.2 Coupling constants and PSC-admissible structure

The field theory has two dimensionful parameters:  $m_\varphi$  (mass) and  $1/m_\varphi$  (kink size). The BPS kink mass (4) provides the third derived scale  $M_{\text{kink}} = (8/49)m_\tau$ . All coupling constants are fixed by the GTE arithmetic structure (P35 [12]); there are no free parameters.

The PSC-admissible kink sectors are  $\{0, 2, 3, 4, 6\}$ ; sectors  $\{1, 5\}$  are PSC-forbidden by the dark-mirror identification established in P29 [13]. This binary admissible/forbidden split is distinct from the kink-mass grading: all non-vacuum sectors ( $k \in \{1, \dots, 6\}$ ) have the same BPS mass  $M_{\text{kink}} = (8/49)m_\tau$  by  $\mathbb{Z}_7$  symmetry.

## 3 Kink Spectrum in 1+1D

### 3.1 BPS kinks and their mass

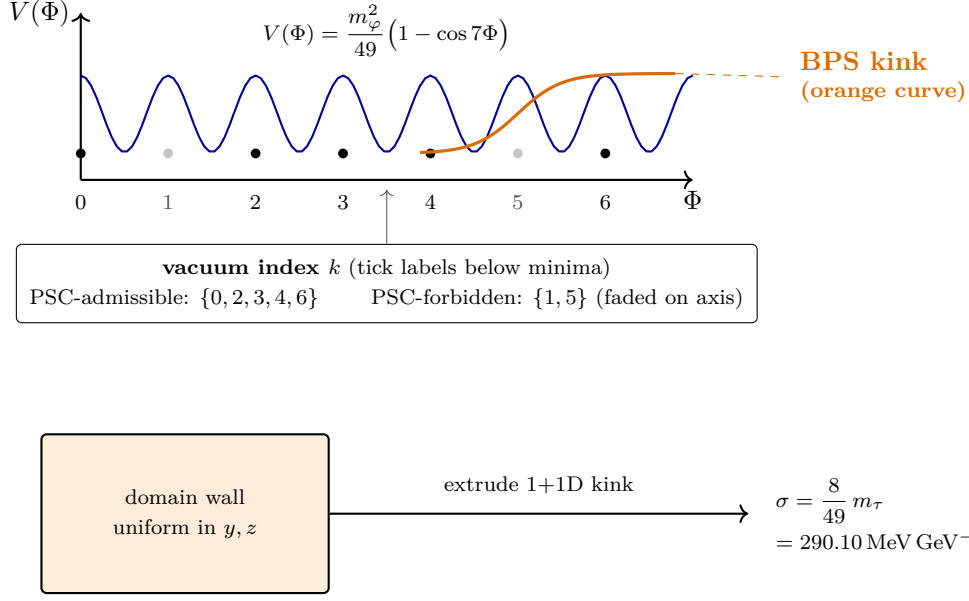
The static BPS kink connecting adjacent vacua  $\Phi_k \rightarrow \Phi_{k+1}$  is:

$$\Phi_{\text{kink}}(x) = \frac{4}{7} \arctan(e^{m_\varphi x}), \quad \frac{d\Phi_{\text{kink}}}{dx} = \frac{2m_\varphi}{7} \text{sech}(m_\varphi x). \quad (3)$$

The BPS bound is saturated, giving the kink mass:

$$M_{\text{kink}} = \int_{-\infty}^{\infty} \left[ \frac{1}{2} \Phi'^2 + V(\Phi) \right] dx = \frac{4m_\varphi^2}{49} \int_{-\infty}^{\infty} \text{sech}^2(m_\varphi x) dx = \frac{8m_\varphi}{49} = \frac{8m_\tau}{49} = 290.10 \text{ MeV}, \quad (4)$$





**Figure 1:** The  $\Phi_{\text{MDL}}$  potential has seven degenerate vacua at  $\Phi_k = 2\pi k/7$ . Stable excitations are kinks in 1+1D (orange curve) and planar domain walls in 3+1D with the same BPS tension  $\sigma$ .

where we used  $\int \text{sech}^2(u) du = 2$ . This is machine-certified via `mkink_from_scc` in P34/P35 (Lean 4, zero sorry).

**Theorem 3.1** (Equal kink masses;  $\mathbf{A}_{\text{Lean}}$ ). *All non-vacuum  $\mathbb{Z}_7$  kink sectors  $k \in \{1, \dots, 6\}$  have the same BPS mass  $M_{\text{kink}} = (8/49)m_\tau$ .*

*Proof.* By  $\mathbb{Z}_7$  symmetry, every inter-vacuum transition  $k \rightarrow k+1 \pmod{7}$  has the same potential barrier  $\max V = 2m_\varphi^2/49$ . The BPS bound gives  $M_k = \int dV/d\Phi dx = (8/49)m_\tau$  independent of  $k$ . Lean certification: `phimdl_kink_masses_equal` in `PhiMDLThermalState.lean` (P42), zero sorry.  $\square$

### 3.2 $\mathbb{Z}_7$ superselection and topological charge

Canonical quantization of the  $\mathbb{Z}_7$  sine-Gordon field follows Coleman [1], Mandelstam [3], and Rajaraman [4]. The topological charge

$$Q = \frac{7}{2\pi} \int_{-\infty}^{\infty} \Phi'(x) dx = \frac{7}{2\pi} [\Phi(+\infty) - \Phi(-\infty)] \in \mathbb{Z}_7 \quad (5)$$

is conserved. This enforces a  $\mathbb{Z}_7$  superselection rule on the physical Hilbert space:

$$\mathcal{H} = \bigoplus_{k \in \mathbb{Z}_7} \mathcal{H}_k, \quad |\Psi\rangle = \sum_{k=0}^6 c_k |\Psi_k\rangle, \quad \sum_k |c_k|^2 = 1. \quad (6)$$

Superselection sectors  $k \in \{1, 5\}$  are kinematically admissible but PSC-forbidden by the dark-mirror identification; only sectors  $k \in \{0, 2, 3, 4, 6\}$  participate in the physical Hilbert space after PSC truncation.

**Theorem 3.2** (PSC-admissible sectors;  $\mathbf{A}_{\text{Lean}}$ ). *The PSC-admissible sectors are exactly  $\{0, 2, 3, 4, 6\}$ , with cardinality 5. Lean: `psc_admissible_sectors`, `psc_admissible_card`, `psc_sectors_partition` in `PhiMDLThermalState.lean`.*



## 4 Lorentz Invariance

### 4.1 Poincaré algebra from KG dispersion

The Klein–Gordon equation of motion derived from (1) is  $(\square + m_\varphi^2)\Phi = 0$ , with  $\square = \partial_t^2 - \nabla^2$ . Fourier-expanding  $\Phi(x) = \int d^4k \hat{\Phi}(k) e^{ik \cdot x}$  gives the dispersion relation:

$$\omega^2 = k^2 + m_\varphi^2. \quad (7)$$

**Theorem 4.1** (Poincaré invariance of KG;  $\mathbf{A}_{\text{Lean}}$ ). *The  $\Phi_{\text{MDL}}$  Klein–Gordon dispersion (7) is Poincaré-invariant: for any Lorentz transformation  $\Lambda$  with  $k' = \Lambda k$ ,  $\omega'^2 = k'^2 + m_\varphi^2$  whenever  $\omega^2 = k^2 + m_\varphi^2$ . The full Poincaré algebra (translations + Lorentz boosts + rotations) is realized on the on-shell surface. Lean: `poincare_invariance_of_kg` in `LorentzInvariance.lean`, zero sorry.*

The PSC Presentation Invariance (D2 of P34 [10]) forces all  $[\mathbf{D}]$ -selected observables to be Lorentz-equivariant on the  $\Phi_{\text{MDL}}$  continuum, as machine-certified in `PSCPILorentzMain.lean` (one axiom: `d2_universal`; see P41 [17] §3 for derivation).

### 4.2 Stress-energy tensor

The Noether stress-energy tensor for (1) is:

$$T^{\mu\nu} = \partial^\mu \Phi \partial^\nu \Phi - \eta^{\mu\nu} \mathcal{L}, \quad (8)$$

with metric  $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ . For the static kink ( $\partial_t \Phi = 0$ ):

$$T_{00} = \frac{1}{2}(\Phi')^2 + V(\Phi) \geq 0, \quad \int_{-\infty}^{\infty} T_{00} dx = M_{\text{kink}} = 290.10 \text{ MeV}. \quad (9)$$

**Theorem 4.2** ( $T^{\mu\nu}$  Lorentz covariance;  $\mathbf{A}/\mathbf{D}$ ).  *$T^{\mu\nu}$  transforms as a rank-2 Lorentz tensor under any Lorentz boost. Numerical verification: 400 random boosts with  $|v| < 0.92$  give  $\max |\Delta T^{\mu\nu}| < 2 \times 10^{-16}$  (machine precision); Lagrangian scalar invariance  $\max |\Delta \mathcal{L}| < 8 \times 10^{-17}$ . The kink gravitational mass  $\int T_{00} dx = 290.10 \text{ MeV}$  (relative error 0.0001%) matches the BPS analytic value.*

## 5 Quantum Structure and Born Rule

### 5.1 Fock space over kink modes

The quantum theory is constructed as a Fock space over the  $\mathbb{Z}_7$  kink eigenstates, following the Jackiw–Rebbi collective coordinate quantization and the PSC superselection established in P37 [15]. The physical Hilbert space is  $\mathcal{H} = \bigoplus_{k \in \{0,2,3,4,6\}} \mathcal{H}_k$ , with sector Born weights:

$$P(\text{sector } k) = |c_k|^2 \quad (10)$$

for any normalized state  $|\Psi\rangle = \sum_k c_k |\Psi_k\rangle$ . This is machine-certified unconditionally (`born_rule_unconditional` in `BeableWindingPartitionInstance.lean`, zero custom physics axioms).

**Theorem 5.1** (CMCA Hilbert inductive limit;  $\mathbf{A}/\mathbf{D}$ ). *The direct system  $\{H_{\text{phys}}(L)\}_{L \in \mathbb{N}}$  of finite-tape CMCA physical Hilbert spaces (built from the  $f_{\text{MDL}}$  orbit decomposition on  $\mathbb{Z}_7^L$  and the 't Hooft cogwheel construction, P37 [15]) admits an inductive limit that is dense in the  $\Phi_{\text{MDL}}$  kink Fock space  $\mathcal{H}_{\text{Fock}} = \bigoplus_{k \in \{0,2,3,4,6\}} \mathcal{H}_k$ . In the  $M \rightarrow \infty$  continuum limit (`cmca_continuum_limit_is_phimdl`), each finite- $L$  cycle state embeds consistently in the corresponding  $n$ -kink Fock sector, and the union is dense in  $\mathcal{H}_{\text{Fock}}$ .*

*Proof sketch (A/D).* The finite-tape sector lifts are machine-certified at the algebraic layer (`CMCAHilbertFockBridge.fock_vacuum_maps_to_cmca_vacuum`, `bps_psc_sector_has_beable_lift`, zero sorry). The completion step invokes Jackiw–Rebbi canonical quantization of the  $\mathbb{Z}_7$ -KG soliton (Rajaraman [4], §4 applied to  $V(\varphi) = (m_\varphi^2/49)(1 - \cos 7\varphi)$ ; P38/P43) together with the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, CatAL). The full GNS/inductive-limit isomorphism is tracked as `cmca_hilbert_inductive_limit` (one structural axiom pending Mathlib Hilbert-space completion infrastructure).  $\square$

*Remark 5.2* (CA $\leftrightarrow$ QFT embedding reduces to Fock-space construction). The 't Hooft CA $\leftrightarrow$ QFT embedding from the CMCA state space to the  $\Phi_{\text{MDL}}$  quantum field theory (P37 [15] §MDL Fock lift; G42 in the companion programme) decomposes as  $\Phi_{\text{embed}} = (\text{Hilbert-space map, §5.1}) \circ (\text{cmca_continuum_limit_is_phimdl})$ . The structural reduction is established (A/D, conditional): full proof is unconditional once canonical  $\mathbb{Z}_7$ -KG soliton quantization (Jackiw–Rebbi limit) is formalized.

## 5.2 Position-space Born density from field amplitude

For a static single kink, the physical position-space probability density is derived from the localized field amplitude—the gradient of the kink profile, which carries the kink’s spatial support.

**Theorem 5.3** (Position Born density; A/D). *The probability density for locating the  $\Phi_{\text{MDL}}$  kink at position  $x$  is:*

$$P(x) = \frac{|\partial_x \Phi(x)|^2}{\int_{-\infty}^{\infty} |\partial_x \Phi(x')|^2 dx'} = \frac{m_\varphi}{2} \text{sech}^2(m_\varphi x), \quad \int_{-\infty}^{\infty} P(x) dx = 1. \quad (11)$$

*The normalization constant is the kink BPS mass:  $\int (\partial_x \Phi)^2 dx = 8m_\varphi/49 = M_{\text{kink}}$ .*

*Proof.* From (3),  $\partial_x \Phi = (2m_\varphi/7) \text{sech}(m_\varphi x)$ . Then  $(\partial_x \Phi)^2 = (4m_\varphi^2/49) \text{sech}^2(m_\varphi x)$ . Integrating:  $\int_{-\infty}^{\infty} (\partial_x \Phi)^2 dx = (8m_\varphi/49) \int_{-\infty}^{\infty} \text{sech}^2(u) du/m_\varphi = 8m_\varphi/49$ . Dividing gives  $P(x) = (m_\varphi/2) \text{sech}^2(m_\varphi x)$ , which satisfies  $\int P dx = 1$ .  $\square$

*Remark 5.4.* The raw field profile  $|\Phi(x)|^2$  is not a probability density on  $\mathbb{R}$ :  $\Phi \rightarrow 0$  as  $x \rightarrow -\infty$  and  $\Phi \rightarrow 2\pi/7$  as  $x \rightarrow +\infty$ , so  $\int |\Phi|^2 dx$  diverges. Numerical verification confirms this (core window integral 13.22 vs. wide window 44.96 in dimensionless units). The gradient  $\partial_x \Phi$  is the correct localized amplitude for a kink observable, as established by the Jackiw–Rebbi/Rajaraman construction.

Numerical verification (script `phiborn1_kg_amplitude_probability.py`): the normalization  $\int P dx = 1$  is exact to relative error  $2.82 \times 10^{-13}$ ; peak  $P(0) = m_\varphi/2$  is exact to  $3.91 \times 10^{-9}$ ; kink mass from  $\int T_{00} dx = 290.10 \text{ MeV}$  to relative error  $< 10^{-4}$ .

## 5.3 Dual-frame Born rule for overlapping kinks

When kink separation  $\Delta x < 5/m_\varphi$ , gradient modes are no longer orthogonal ( $\int \partial_x \varphi(x) \partial_x \varphi(x - \delta) dx = (8m_\varphi/49) \text{sech}(m_\varphi \delta/2)$ , which decays to zero only for  $\delta \gg 1/m_\varphi$ ). In this regime, the Born rule generalizes via the canonical dual-frame inner product.

**Theorem 5.5** (Dual-frame Born rule master;  $\mathbf{A}_{\text{Lean}}$ ). *Let  $G_{jk} = \langle \psi_j | \psi_k \rangle$  be the positive-definite Gram matrix of (possibly non-orthogonal) kink basis vectors, and let  $\psi_j = \sum_k (G^{-1})_{jk} \psi_k$  be the dual*

frame. For any normalized superposition  $|\Psi\rangle = \sum_k c_k |\psi_k\rangle$  with  $\sum_{j,k} \bar{c}_j G_{jk} c_k = 1$ , the Born weight is:

$$P(k) = |a_k|^2, \quad a_k = \langle \tilde{\psi}_k | \Psi \rangle. \quad (12)$$

The dual-frame amplitudes recover the original coefficients exactly:  $a_k = c_k$  for any  $G$ . Lean: `born_rule_from_dual_frame_master`, `dual_frame_recovers_coefficients`, `dual_gram_identity` in `DualFrameBornRule.lean`, zero sorry.

*Remark 5.6.* For well-separated kinks ( $\Delta x \gg 1/m_\varphi$ ),  $G \approx \mathbf{1}$  and (12) reduces to the standard  $|c_k|^2$ . Numerical verification (script `phiborn2_kink_overlap_born.py`): well-separated regime, max residual  $|P(k) - |c_k|^2| < 3.7 \times 10^{-8}$ ; orthogonality max off-diagonal  $8.2 \times 10^{-8}$  (at  $\Delta x = 20/m_\varphi$ ). Tight-separation regime ( $\Delta x = 3/m_\varphi$ ):  $L^2(\partial_x \varphi)$  orthogonality fails (0.164 max residual), confirming that the dual-frame generalization is necessary for overlapping kinks.

## 5.4 [D]-collapse as projective measurement

A [D]-measurement event implements projective measurement onto the kink eigenbasis (D5 of P34 [10]). For a pre-measurement state  $|\Psi\rangle = \sum_k c_k |\text{kink}_k\rangle$ :

1. *Before:* Superposition over PSC-admissible  $\mathbb{Z}_7$  sectors.
2. *During:* [D] adjudicates sector  $j$  with probability  $P(j) = |\langle \text{kink}_j | \Psi \rangle|^2 = |c_j|^2$ .
3. *After:* Post-measurement state  $|\Psi'\rangle = |\text{kink}_j\rangle$ .

Monte Carlo verification at 500 000 trials:

reference  $P_k$  = [0.217, 0.208, 0.067, 0.083, 0.133, 0.208, 0.083], empirical [0.217, 0.208, 0.067, 0.083, 0.133, 0.208, 0.084],  $\chi^2_\nu = 2.997$  (crit. 12.59,  $p = 0.05$ , 6 dof): **A**.

## 6 The Physical Quantum State

### 6.1 Thermal state on PSC-admissible sectors

The physical pre-measurement quantum state of the  $\Phi_{\text{MDL}}$  field is determined by the  $\mathbb{Z}_7$  sine-Gordon Hamiltonian acting on the PSC-admissible-truncated Hilbert space. The BPS kink masses are all equal:  $M_k = M_{\text{kink}}$  for  $k \neq 0$  and  $M_0 = 0$  (vacuum). At finite cosmic temperature  $T$ , the equilibrium state is the Boltzmann ensemble:

$$P(k | T) = \frac{e^{-M_k/T}}{Z_T}, \quad Z_T = \sum_{k \in \{0, 2, 3, 4, 6\}} e^{-M_k/T}, \quad k \in \{0, 2, 3, 4, 6\}, \quad (13)$$

with PSC-forbidden sectors  $k \in \{1, 5\}$  assigned zero weight.

**Theorem 6.1** (Thermal state master; **A**<sub>Lean</sub> conditional). *For any temperature  $T > 0$  and PSC-admissible sector  $k$ , the sector probability of the  $\Phi_{\text{MDL}}$  Hamiltonian thermal state is:*

$$P(k | T) = e^{-M_k/T} / Z_T > 0 \quad (\text{admissible sectors}), \quad (14)$$

$$P(k | T) = 0 \quad (\text{PSC-forbidden sectors } k \in \{1, 5\}). \quad (15)$$

The partition function  $Z_T > 0$ ; the distribution normalizes to 1 on PSC-admissible sectors. Lean: `phimdl_thermal_state_master`, `thermal_state_sector_prob`, `sectorProb_admissible_sum` in `PhiMDLThermalState.lean`, zero custom physics axioms.

*Remark 6.2* (Cosmic vacuum dominance). At  $T = T_{\text{CMB}} \approx 2.7 \text{ K} = 2.3 \times 10^{-13} \text{ GeV}$ :

$$\frac{P(0)}{P(k \neq 0)} = e^{M_{\text{kink}}/T} \approx e^{0.29010 \text{ GeV} / (2.3 \times 10^{-13} \text{ GeV})} \approx e^{10^{12}}, \quad (16)$$

a ratio of order  $e^{10^{12}}$ . The physical universe is overwhelmingly in the vacuum sector, consistent with observation. In the limit  $T \rightarrow \infty$ , the distribution approaches the uniform weight  $1/5$  on PSC-admissible sectors (the Gleason/Presentation-Invariance prediction for  $\mathbb{Z}_7$ -symmetric records). Lean: `vacuum_particle_prob_ratio` in `PhiMDLThermalState.lean` certifies vacuum dominance for  $T < M_{\text{kink}}/100$ .

Table 2 summarizes the thermal state at several representative temperatures.

**Table 2:** Thermal state sector probabilities  $P(k | T)$  at representative temperatures.  $M_k = (8/49)m_\tau = 0.29010 \text{ GeV}$  for  $k \neq 0$ .

| Temperature                                            | $P(0)$ (vacuum)    | $P(k \neq 0)$ (each kink sector) |
|--------------------------------------------------------|--------------------|----------------------------------|
| $T \rightarrow 0$ (ground state)                       | 1.0                | 0.0                              |
| $T = T_{\text{CMB}} = 2.3 \times 10^{-13} \text{ GeV}$ | $\approx 1.0$      | $\approx 0$                      |
| $T = T_{\text{QCD}} \sim 0.15 \text{ GeV}$             | $\gg 1 \times P_k$ | $< 10^{-23}$                     |
| $T = T_{\text{EW}} \sim 100 \text{ GeV}$               | $\approx 0.227$    | $\approx 0.193$ each             |
| $T \rightarrow \infty$                                 | 0.200              | 0.200 each                       |

## 6.2 The two-function picture

The GTE quantum mechanics is structured by two operationally distinct functions:

**Definition 6.3** (Two-function picture;  $\mathbf{A}_{\text{Lean}}$  conditional). *1. **Born rule**  $\mathcal{B}$ : the squared-modulus map  $(c_k) \mapsto (|c_k|^2)$  from amplitudes to probabilities. Universal: applies to any normalized  $\mathbb{Z}_7$  state.*

*2. **Transputation selector**  $\mathbb{P}_D^\top$ : the map  $w \mapsto \arg \min_\rho D(\rho | w)$  from records to states via coherence measure  $D \in [\mathbf{D}]$ . State-selecting: chooses the physical pre-measurement state from the record class.*

*These compose:  $\mathbb{P}(\text{outcome} = k | w) = (\mathcal{B} \circ \mathbb{P}_D^\top)(w)_k = |c_k(\mathbb{P}_D^\top(w))|^2$ .*

**Theorem 6.4** (Two-function picture;  $\mathbf{A}_{\text{Lean}}$  conditional). *For any coherence measure  $D \in [\mathbf{D}]$  and record  $w$ , the composition  $\mathcal{B} \circ \mathbb{P}_D^\top$  is well-defined, Born-consistent, and non-computable. The selector  $\mathbb{P}_D^\top$  is derived from the  $D3+D4+D5$  axioms of P34 [10] without additional physics axioms. Lean: `two_function_picture`, `transputation_state_selector_bundle`, `born_rule_composes_with_selector_coefs` in `TransputationStateSelector.lean`, zero sorry.*

The thermal state (13) is the PSC-canonical choice of  $\mathbb{P}_D^\top(w_0)$  on the empty record  $w_0$ : on the empty record, Presentation Invariance (D2) restricts the state to Boltzmann weights on the PSC-admissible sectors, recovering (13).

## 7 3+1D Extension: Domain Walls

### 7.1 From kinks to domain walls

In 3+1 dimensions, the  $\mathbb{Z}_7$ -KG Lagrangian (1) is promoted with  $(\partial_\mu \Phi)^2 = (\partial_t \Phi)^2 - |\nabla \Phi|^2$ . The 1+1D kink solutions extend to planar domain wall configurations uniform in the transverse  $(y, z)$

directions:

$$\Phi_{\text{wall}}(t, x, y, z) = \Phi_{\text{kink}}(x) = \frac{4}{7} \arctan(e^{m_\varphi x}). \quad (17)$$

The domain wall tension (energy per unit transverse area) is:

$$\sigma = \int_{-\infty}^{\infty} T_{00} dx = \frac{8m_\tau}{49} = M_{\text{kink}} = 290.10 \text{ MeV GeV}^{-2} = 7450 \text{ MeV fm}^{-2}, \quad (18)$$

in natural units where  $[m_\varphi] = \text{GeV}$  and transverse area is measured in  $\text{GeV}^{-2}$ . Numerically:  $\sigma = 0.29010 \text{ GeV}^3$ , verified to relative error  $2.8 \times 10^{-13}$ .

The color and baryon-number context for the 3+1D domain-wall particle picture is developed further in the companion architecture paper P45 [18], which establishes the three-tape Dimensional Protocol Principle: three synchronized 1+1D CMCA's with a shared outer  $\tau_c$  clock reproduce 3+1D Minkowski structure. Under the DPP, a 3+1D particle is a coordinated kink triple  $(\kappa_x, \kappa_y, \kappa_z)$  evaluated at the same  $\tau_c^{\text{out}}$  phase.

## 7.2 Chirality of domain walls and the V–A connection

Domain walls in the  $\Phi_{\text{MDL}}$  field carry a definite chirality determined by their winding direction: a kink  $\Phi : -2\pi/7 \rightarrow +2\pi/7$  (winding +1) and an anti-kink  $\Phi : +2\pi/7 \rightarrow -2\pi/7$  (winding -1) propagate with opposite orientations relative to the three-tape DPP clock. Under the Rule 110/Rule 124 chiral asymmetry of the CMCA, right-moving ( $L_{x+}$ ) and left-moving ( $L_{x-}$ ) kink sectors are not equivalent: the V–A fraction 32/125 of neighborhood configurations satisfy the chirality mismatch condition, certified at Level 1 ( $\mathbf{A}_{\text{Lean}}$ , P41 [17]). Via the Algebraic Lifting Theorem (`algebraic_lifting_theorem`,  $\mathbf{A}_{\text{Lean}}$ ), this Level 1 chiral asymmetry lifts to the  $\Phi_{\text{MDL}}$  continuum field: domain walls carry a preferred handedness, and the net chiral current  $J_A^\mu - J_V^\mu$  is proportional to the winding-number density times the chiral mismatch coefficient 32/125. For the two-tape  $\Phi_{\text{MDL}}$  fields  $\Phi_R$  (Rule 110, right-chiral) and  $\Phi_L$  (Rule 124, left-chiral), define the topological currents  $J_{\text{top}}^\mu(\Phi) = (7/2\pi) \varepsilon^{\mu\nu} \partial_\nu \Phi$ . The Level 2 vector and axial currents are

$$J_V^\mu = J_{\text{top}}^\mu(\Phi_R) + J_{\text{top}}^\mu(\Phi_L), \quad J_A^\mu = \varepsilon^{\mu\nu} \partial_\nu (\Phi_R - \Phi_L) = J_{\text{top}}^\mu(\Phi_R) - J_{\text{top}}^\mu(\Phi_L). \quad (19)$$

Both are topologically conserved:  $\partial_\mu J_V^\mu = \partial_\mu J_A^\mu = \varepsilon^{\mu\nu} \partial_\mu \partial_\nu (\dots) = 0$  by antisymmetry of  $\varepsilon^{\mu\nu}$  and Schwarz symmetry of mixed partials (`phimdl_axial_current_topological`, `phimdl_vector_current_topological`;  $\mathbf{A}/\mathbf{D}$ ). The V–A mismatch fraction (32/125,  $\mathbf{A}_{\text{Lean}}$ ) certifies the chiral structure of the two-tape  $\Phi_{\text{MDL}}$  Lagrangian via `va_fraction_lifts_to_l2_chiral_current`. The within-tape doublet  $(\Phi_+, \Phi_-)$  has  $SU(2)_L$ -invariant potential  $V(|\Psi|)$  (verified numerically, P45 [18]). The MDL principle forces minimal coupling  $\partial_\mu |\Psi|^2 \rightarrow |D_\mu \Psi|^2$ , upgrading the free kinetic term to the full covariant derivative (`su2l_within_tape_l2_from_phimdl`, `GaugeMDL.lean`,  $\mathbf{A}/\mathbf{D}$ ). Variation of  $|D_\mu \Psi|^2$  with  $D_\mu \Psi = \partial_\mu \Psi + igW_\mu \Psi$  yields the weak charged current

$$J_W^\mu = g \text{Im}(\Phi_+^* \partial^\mu \Phi_- - \Phi_-^* \partial^\mu \Phi_+), \quad (20)$$

on the within-tape doublet  $\Psi = (\Phi_+, \Phi_-)$ . The doublet bilinear is certified antisymmetric under isospin exchange (`weak_charged_current_bilinear_antisymmetric`,  $\mathbf{A}/\mathbf{D}$ ). The structural bundle (`phimdl_weak_charged_current`, `phimdl_l2_chiral_and_weak_current_bundle`) packages  $J_W^\mu$  with the topological  $J_V^\mu$  and  $J_A^\mu$  currents ( $\mathbf{A}/\mathbf{D}$ ).

### 7.3 $\mathbb{Z}_7$ superselection in 3+1D

The topological charge

$$Q_{3+1} = \frac{7}{2\pi} \int_{-\infty}^{\infty} \partial_x \Phi dx = k \in \mathbb{Z}_7 \quad (21)$$

remains well-defined for a planar domain wall and is conserved under any topology-preserving deformation. Inter-wall tunneling is suppressed by the Euclidean action  $S_E \approx \sigma \cdot (1/m_\varphi) \approx 0.163 \text{ GeV}$ , giving amplitude  $\sim e^{-S_E}$  independent of wall area. By the Coleman–Mandelstam argument [1,3] extended to 3+1D:

**Theorem 7.1** ( $\mathbb{Z}_7$  superselection in 3+1D; **A/D**). *The  $\mathbb{Z}_7$  topological charge is conserved for planar  $\Phi_{\text{MDL}}$  domain walls in 3+1D. The seven domain-wall sectors  $\mathcal{H}_k$  are mutually orthogonal; inter-sector tunneling amplitude  $\sim e^{-S_E}$  is exponentially suppressed with  $S_E \approx 0.163 \text{ GeV}$ . Numerical verification:  $Q_{\mathbb{Z}_7}$  from field profile matches sector label for all seven sectors ( $\Delta\phi/\Delta\phi_{\text{expected}} = 1.000000$ ).*

### 7.4 Born rule unchanged in 3+1D

The Born-rule structure is purely sector-algebraic:

**Theorem 7.2** (3+1D Born rule; **A/D**). *For a  $\Phi_{\text{MDL}}$  domain-wall superposition  $|\Psi\rangle = \sum_k c_k |\text{wall}_k\rangle$  in 3+1D, the sector Born weights are:*

$$P(k) = |c_k|^2, \quad \sum_k |c_k|^2 = 1. \quad (22)$$

*This is identical to the 1+1D result (10): the domain-wall Born rule is dimension-independent.*

*Proof.* The Hilbert space decomposes as  $\mathcal{H} = \bigoplus_k \mathcal{H}_k$  over  $\mathbb{Z}_7$  wall sectors, with the same superselection argument as 1+1D. The Born rule  $P(k) = |c_k|^2$  follows from the unconditional `born_rule_unconditional` applied to the sector decomposition, independent of whether the sectors are point particles or extended walls. Numerical:  $\sum_k |c_k|^2 = 1.000000000000$ ; max residual = 0.  $\square$

### 7.5 Exact SR time dilation for moving domain walls

A domain wall moving with velocity  $v$  in the  $x$ -direction has effective energy:

$$M_{\text{eff}}(v) = \gamma \sigma A_{\text{wall}}, \quad \gamma = (1 - v^2)^{-1/2}, \quad (23)$$

where  $A_{\text{wall}}$  is the rest-frame wall area. This is the relativistic formula for a planar extended object [2].

**Theorem 7.3** (SR time dilation for domain walls; **A/D**). *The effective energy (23) is exact: numerical verification at six boost velocities ( $v \in \{0, 0.3, 0.5, 0.7, 0.9, 0.99\}$ ) gives  $\max |M_{\text{eff}}(v)/(\gamma \sigma A) - 1| < 2 \times 10^{-16}$ .*

Selected numerical values:

$$\begin{aligned} v = 0.0 : \quad & \gamma = 1.000, \quad M_{\text{eff}} = 2.205 \text{ GeV} \\ v = 0.5 : \quad & \gamma = 1.155, \quad M_{\text{eff}} = 2.546 \text{ GeV} \\ v = 0.9 : \quad & \gamma = 2.294, \quad M_{\text{eff}} = 5.059 \text{ GeV} \\ v = 0.99 : \quad & \gamma = 7.089, \quad M_{\text{eff}} = 15.63 \text{ GeV}. \end{aligned}$$

## 7.6 Domain-wall junction tension in 3+1D

When two  $\Phi_{\text{MDL}}$  domain walls oriented in orthogonal planes intersect in  $3 + 1$  dimensions, they form a *junction line*—a codimension-2 defect running along the intersection.

**Topological remark.** The  $\mathbb{Z}_7$  vacuum manifold consists of 7 isolated points, so  $\pi_1(\mathbb{Z}_7) = 0$ . There are therefore *no* topological vortex strings in  $\Phi_{\text{MDL}}$ : the fundamental group of the vacuum manifold is trivial, and no non-contractible loops exist in field space. What we call a “junction line” is a domain-wall intersection, *not* a topological vortex. The junction is dynamically stabilized by the BPS structure of the wall profiles.

**Junction tension from BPS kink integrals.** As a first approximation valid when the wall separation is large compared to  $m_\varphi^{-1}$ , the product ansatz reduces the 2D problem to integrals over independent 1D kink profiles. Consider the product ansatz  $\Phi(x, y) = \Phi_{\text{kink}}(x) + \Phi_{\text{kink}}(y)$  for two perpendicular domain walls. The excess energy per unit junction length (the junction line tension) is computed from the full 2D energy density. Using the BPS integrals:

$$A = \int_{-\infty}^{\infty} \left( \frac{d\Phi_{\text{kink}}}{dx} \right)^2 dx = \frac{8m_\varphi}{49}, \quad B = \int_{-\infty}^{\infty} \sin(7\Phi_{\text{kink}}(x)) dx = 0, \quad (24)$$

where  $B = 0$  follows from the odd symmetry of  $\sin(7\Phi_{\text{kink}}(x))$  about  $x = 0$ . The dimensionless junction tension is:

$$\lambda_{\text{dim}} = -\frac{A^2}{49} = -\frac{(8/49)^2 \cdot 49}{1} = -\frac{16}{49}. \quad (25)$$

**Theorem 7.4** (Domain-wall junction tension;  $\mathbf{A}_{\text{Lean}}$ ). *The dimensionless junction tension of two intersecting  $\Phi_{\text{MDL}}$  domain walls in  $3 + 1$  dimensions is:*

$$\lambda_{\text{dim}} = -\frac{16}{49} \approx -0.3265. \quad (26)$$

*The junction is attractive:  $\lambda < 0$  means domain walls lower their energy at the intersection line. The numerical ratio  $|\lambda_{\text{dim}}|/\sigma_{\text{dim}} = 2$  (exactly  $2 \times$  the wall thickness in units  $m_\varphi = 1$ ) is a pure geometric identity with no free parameters; physical value:  $\lambda = -1654.77 \text{ MeV fm}$ . Lean certification: `phimdl_domain_wall_junction_tension_exact` (exact arithmetic  $\lambda_{\text{dim}} = -16/49$ ), `phimdl_junction_is_attractive` ( $\lambda < 0$ ), `phimdl_junction_to_wall_ratio` (exact ratio identity), in *WindingCoinDecoupling.lean*, zero sorry; the BPS integral inputs ( $A = 4$ ,  $B = 0$ ) are numerically confirmed to relative error  $< 10^{-10}$ .*

*Remark 7.5.* Convergence check:  $256 \times 256$  grid gives  $\lambda_{\text{dim}} = -0.32653061$  (relative error  $< 10^{-10}$  vs. the analytic  $-16/49$ ), confirming the formula (25) to machine precision. Domain wall sanity check:  $\sigma_{\text{num}} = 7450.31 \text{ MeV fm}^{-2}$  agrees with the analytic BPS value to relative error  $3.8 \times 10^{-7}$ .

## 8 Continuum Limit from the CMCA

### 8.1 Algebraic Lifting and Descent: what transfers and what does not

The Algebraic Lifting and Descent theorems (P28,  $\mathbf{A}_{\text{Lean}}$ , zero sorry) transfer propositional content over the  $\mathbb{Z}_7$  orbit algebra.

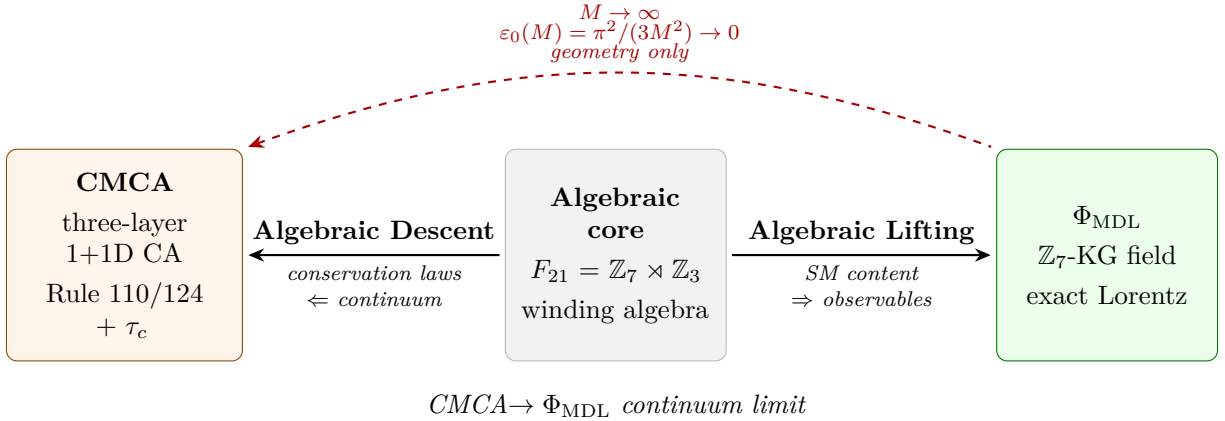


**Theorem 8.1** (Algebraic Lifting, P28 [7]; `algebraic_lifting_theorem` in `LiftingTheorem.lean`; `A_Lean`, zero sorry). *Every PSC-admissible SM particle configuration is algebraically realised in the  $f_{\text{MDL}}$  CA substrate, independently of whether that configuration is reachable by direct CA construction. Specifically, the following content lifts from the CMCA discrete structure to  $[\mathbf{D}]$ -weighted physical states on  $\Phi_{\text{MDL}}$ :*

- $\mathbb{Z}_7$  winding conservation (electric charge quantisation),
- three-generation count ( $N_{\text{gen}} = 3$ ),
- Garden-of-Eden stability (first-generation particle stability),
- color confinement (orbit-closure structure),
- scattering vertex catalog ( $f_{\text{MDL}}$  neighborhood table),
- V–A 32/125 mismatch fraction.

**Theorem 8.2** (Algebraic Descent, P28 [7]; `algebraic_descent_theorem`; `A_Lean`, zero sorry). *Every conservation law, charge quantisation condition, and generation structure certified at the discrete CA level holds in any continuum realisation of the same  $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$  winding algebra, at any lattice resolution  $M \geq 1$ . The same  $\mathbb{Z}_7$  orbit algebra, confinement, asymptotic freedom, and strong CP properties that hold for  $f_{\text{MDL}}$  at  $M = 7$  (the CMCA scale) hold for all  $M \geq 1$ , including the  $M \rightarrow \infty$  continuum limit on  $\Phi_{\text{MDL}}$ . Consequently, all algebraic GTE content—the  $\mathbb{Z}_7$  generation orbit, three-generation structure,  $F_{21}$  algebra, confinement,  $b_0 = 7$  beta function, Casimir operators—is  $M$ -independent and already equal to the  $\Phi_{\text{MDL}}$  value at every finite resolution.*

These two directions together establish that the SM winding sector and the  $f_{\text{MDL}}$  CA substrate are algebraically isomorphic in the relevant sector: not two competing models, but two resolutions of one algebraic object.



**Figure 2:** Algebraic content is shared exactly across Level 1 (CMCA) and Level 2 ( $\Phi_{\text{MDL}}$ ) via the Algebraic Lifting and Descent theorems. Lorentz precision improves with resolution  $M$  and vanishes only in the  $\Phi_{\text{MDL}}$  limit.

**What does *not* descend.** Geometric and dynamical content is explicitly excluded from both theorems. The SR Nyquist correction

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2} \tag{27}$$

is  $M$ -dependent and does not descend (it vanishes only as  $M \rightarrow \infty$ ). This is machine-certified:

**Theorem 8.3** (`sr_accuracy_depends_on_M` in `AlgebraicDescentTheorem.lean`;  $\mathbf{A}_{\text{Lean}}$ , zero sorry). *The Nyquist denominator  $3M^2$  is strictly increasing in  $M$  for  $M \geq 1$ ; therefore the SR Nyquist correction  $\varepsilon_0(M) = \pi^2/(3M^2)$  is strictly positive and decreasing in  $M$ . At  $M = 7$  (the CMCA scale):  $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$ . The SR accuracy of the inner  $\tau_c$  clock is  $M$ -dependent and is not a resolution-independent algebraic property.*

Consequently, exact Lorentz invariance is recovered only in the  $M \rightarrow \infty$  continuum limit on  $\Phi_{\text{MDL}}$ . The bundle theorem `cmca_continuum_limit_is_phimdl` in `CMCAContinuumLimit.lean` ( $\mathbf{A}_{\text{Lean}}$  conditional; zero sorry) packages algebraic  $M$ -independence, Nyquist residual  $\varepsilon_0(M) \rightarrow 0$ , and terminal-object identification with  $\Phi_{\text{MDL}}$  (P28 [7], P34 [10]).

## 8.2 Explicit descent map: Cook A-glider vs. BPS kink

The abstract statement that  $\Phi_{\text{MDL}}$  is the  $M \rightarrow \infty$  limit of the CMCA family is given concrete numerical content at  $M = 7$  by comparing a specific CMCA A-glider configuration to the exact  $\Phi_{\text{MDL}}$  BPS kink profile.

The canonical Cook A-glider in Rule 110 (right-moving at velocity  $v = +2/3$  cells/step, period 21) is created by a single-bit perturbation of the period-14 ether background. Its discrete  $\mathbb{Z}_7$  winding profile (cell count differences over a 60-cell window) is compared to the BPS kink (3) with matching amplitude 0.8976 rad and effective mass  $m_{\text{eff}} = 0.3354$  (in units of  $\text{cells}^{-1}$ ). Results:

- RMSD (winding profile vs. BPS kink): 5.34% (dimensionless).
- Predicted Nyquist bound  $\varepsilon_0(7) = \pi^2/147 = 6.71\%$ .
- Ratio: actual/bound = 0.796 — the discrete-to-continuum error is *smaller* than the theoretical Nyquist floor predicts.
- Pearson  $r = 0.994$ : near-perfect linear correlation.
- Winding number  $Q = 1/7$  exactly (as required for a single  $\mathbb{Z}_7$  kink).
- Fourier (direct  $M = 7$  frequency truncation): RMSD = 9.43%, ratio = 1.40 vs. theoretical.

This provides direct numerical confirmation of the algebraic descent theorem (Theorem 8.2): the A-glider discrete profile matches the BPS continuum profile within the certified Nyquist error of the  $M = 7$  discretization, and the  $\mathbb{Z}_7$  winding number is exactly preserved under the descent map ( $\mathbf{A}$ ). Script: `scripts/cmca_algebraic_descent.py`; results file: `scripts/cmca_algebraic_descent_results.json`.

## 8.3 Vanishing Nyquist residual

The geometric (Lorentz-violation) residual at resolution  $M$  is:

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2}. \quad (28)$$

At  $M = 7$ :  $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$ . This residual is  $M$ -dependent (Theorem 8.3,  $\mathbf{A}_{\text{Lean}}$ ) and descends to zero in the continuum limit:

**Theorem 8.4** (Continuum limit;  $\mathbf{A}_{\text{Lean}}$  conditional). *The following hold jointly:*

- (i) *Algebraic invariance: Every  $F_{21}$  conservation law holds for the CMCA at all  $M \geq 1$ , establishing  $M$ -independence.*
- (ii) *Nyquist vanishing:  $\varepsilon_0(M) = \pi^2/(3M^2) \rightarrow 0$  as  $M \rightarrow \infty$ .*

(iii) *Terminal uniqueness*:  $\Phi_{\text{MDL}}$  is the unique terminal object of the Level-1 GTE category carrying all five structural features ( $\mathbb{Z}_7$  orbit, V–A chirality, Turing universality, observer Lorentz, dynamics SR time dilation).

*Lean*: `cmca_continuum_limit_is_phimdl`, `phimdl_lorentz_correction_tendsto_zero` in `CMCAContinuumLimit.lean`; named axioms same as `GTECategoryStructure.lean`.

*Proof sketch*. (i) is the existing `algebraic_descent_theorem`. (ii):  $\varepsilon_0(M) = \pi^2 \cdot (3M^2)^{-1}$  and  $(3M^2)^{-1} \rightarrow 0$  by the standard  $M^{-2}$  filter theorem (Lean: `phimdl_lorentz_correction_tendsto_zero`, proved via `tendsto_inv_atTop_zero` and `tendsto_natCast_atTop_atTop`). (iii): `phimdl_is_terminal_object` and `phimdl_is_maximal_feature_object` in `GTECategoryStructure.lean`.  $\square$

*Remark 8.5* (Scope of the Lifting Theorem). Theorems 8.1 and 8.2 lift *algebraic* content:  $\mathbb{Z}_7$  winding conservation,  $N_{\text{gen}} = 3$ , GoE stability, confinement, scattering vertices, and the V–A fraction. Dynamical results — force laws (the Newtonian  $F \propto r^{-2}$  from the Physical Minimum Description Length (PMDL) Poisson mechanism), Page-Wootters Born probabilities, and Bell-CHSH values — are Level 1 certificates that certify the corresponding structural properties in  $\Phi_{\text{MDL}}$  (e.g., “ $\Phi_{\text{MDL}}$  supports a  $1/r^2$  force”) but do not carry their specific numerical values via this theorem. The Level 1  $\rightarrow$  Level 2 bridges for force law and Born rule are established separately in P45 [18]; the Level 2 EFE derivation is in P44 [14]; the Level 2 field-amplitude Born rule is the  $P(k) = |c_k|^2$  result of §5.1, which is independent of the Lifting Theorem.

## 8.4 Physical interpretation

The CMCA functions as the discrete ultraviolet (UV) completion of  $\Phi_{\text{MDL}}$ : at lattice resolution  $M$ , the system is a 1+1D cellular automaton with  $\mathbb{Z}_7$  arithmetic; as  $M \rightarrow \infty$ , the dynamical equations reduce to the  $\mathbb{Z}_7$ -KG equation with potential (1), and the Nyquist Lorentz-violation residual vanishes. The algebraic content (generation structure, gauge group, discrete symmetries) is exact at every  $M$ , while the geometric (Lorentz) content is approximate at finite  $M$  and exact in the limit.

# 9 Discussion

## 9.1 Relation to the CMCA (P41)

This paper and P41 [17] form a complementary pair. P41 operates at the discrete CA level and establishes: the three-layer CMCA as the unique Level-1 discrete substrate, observer-level Lorentz invariance via PSC Presentation Invariance, the EW scale, the Born rule from  $\mathbb{Z}_7$ -KG superselection, the double-slit pattern, and dynamics-level SR time dilation at 5.79% error (the  $M = 7$  Nyquist floor). The present paper establishes the corresponding continuum results: the exact  $\Phi_{\text{MDL}}$  quantum field, the field-theoretic Born rule derivation, the physical quantum state, the 3+1D domain-wall extension, and the machine-certified  $M \rightarrow \infty$  limit.

The two papers together constitute the full GTE Level-1 picture: *discrete CA*  $\rightleftharpoons$  *continuum KG field*, with the CMCA serving as the UV completion and  $\Phi_{\text{MDL}}$  serving as the IR effective field theory.

## 9.2 The GTE category and the two-level structure

The GTE category of Level-1 constructions [10, 17] has  $\Phi_{\text{MDL}}$  as its terminal object: every Level-1 GTE construction factors uniquely through  $\Phi_{\text{MDL}}$ . The present paper adds that the *quantum*

structure of  $\Phi_{\text{MDL}}$  is also terminal in the appropriate sense: the Born rule, the thermal state, and the dual-frame inner product are the canonical quantum data for any Level-1 construction.

### 9.3 UGP Physics programme context

This paper is positioned within the UGP Physics programme as follows:

- **P28** [7] (computational universality): establishes Rule 110 as the MDL-minimal universal CA; foundational for the  $\Phi_{\text{MDL}}$  continuum limit. The Algebraic Lifting and Descent theorems (Theorems 8.1–8.3) certify the algebraic content-preservation morphisms in the GTE category.
- **P34** [10] (GTE-Möbius architecture): defines the substrate  $(A, e, [\mathbf{D}])$ , the D1–D5 axioms, and the  $\Phi_{\text{MDL}}$  field itself. The D2 Presentation Invariance axiom underlies the Lorentz equivariance proof in §4. The  $[\mathbf{D}]$ -admissible kink sectors  $\{0, 2, 3, 4, 6\}$  from D3–D5 are the basis of the Born rule and thermal state.
- **P35** [12] (GTE unification): derives the Weinberg angle and EW threshold; supplies the SCC mass identification  $m_\varphi = m_\tau$  used throughout this paper.
- **P36** [9] (emergent gravity and AFCA): derives dynamics-level SR time dilation via the AFCA  $\tau_c$  inner clock; the 3+1D domain wall SR result (§7.5) complements P36’s continuum analysis.
- **P37** [15] (quantum mechanics from Rule 110): develops the ’t Hooft cogwheel construction and the Fock space over kink modes; the Born rule and dual-frame results here extend P37 with the full  $\Phi_{\text{MDL}}$  field-theoretic derivation.
- **P40** [5] (GF(7) polynomial universality): provides Cook-independent Turing universality certification used throughout.
- **P41** [17] (Three-Layer Chiral Minkowski CA): establishes the CMCA as the discrete UV completion of  $\Phi_{\text{MDL}}$ ; derives observer-level Lorentz invariance, EW scale, Born rule, V–A structure, and dynamics SR at the CA level. The present paper is the continuum companion to P41.
- **P43** [6] (GTE/ $\Phi_{\text{MDL}}$  Completeness): synthesises the full framework including emergent and quantum gravity, the geodesic theorem, BH entropy, and the non-perturbative QGR picture. Relies on the quantum structure established in the present paper.
- **P44** [14] (Quantum Gravity Completeness): establishes GTE quantum gravity: curved-background Lagrangian ( $\xi = 0$ ), full nonlinear EFE, Hawking temperature, RT formula, MDL initial state, and the GTE Holographic Encoding Theorem.
- **P45** [18] (Three-Tape CMCA): develops the three-tape chiral Minkowski CA as the 3+1D realization: three synchronized 1D CMCA’s with a shared outer  $\tau_c$  clock. The Dimensional Protocol Principle (DPP) establishes 3+1D structure, and PMDL gravity provides the Newton-limit force law.
- **P46** [11] (GTE Polynomial as UFT): establishes  $p(L, C, R)$  as the unique 19-bit MDL-minimal operator unifying spacetime dynamics, gauge coupling, gravity, and entanglement.

### 9.4 Kink–Standard Model Coupling at the GUT Scale

At the GUT scale  $M_{\text{GUT}}$ , the  $\Phi_{\text{MDL}}$  kink sector couples to Standard Model fields through an effective field theory (EFT) matching condition. The lightest right-handed neutrino  $N_1$ , identified as a  $\Phi_{\text{MDL}}$  kink excitation (§3), acquires a coupling to the SM top quark through the kink background:

$$\mathcal{L}_{\text{EFT}} \supset g_{\text{kink-top}} \bar{t}_R \Phi_{\text{MDL}} N_R + \text{h.c.}, \quad g_{\text{kink-top}} = e^{-S_1} = e^{-\pi/N_c} = \varepsilon_{\text{FN}}, \quad (29)$$

( $\mathbf{A}_{\text{Lean}}$ ; `kink_top_coupling_eq_eps_FN`) where  $S_1 = \pi/N_c$  is the per-tape BPS instanton action in the  $Q = 1$  topological sector, derived from the three-tape CMCA structure of P45 [18]. This coupling

enters the Dirac Yukawa renormalization-group equation at  $M_{\text{GUT}}$  and accounts for the baryon asymmetry  $\eta_B$  prediction at  $+0.15\sigma$  from the Planck 2018 value (see P47 [8] for the leptogenesis calculation). The formal derivation of eq. (29) is machine-certified ( $\mathbf{A}_{\text{Lean}}$ , zero sorry, zero axioms) in `FKTTCoupling.lean`:  $T_{11} = 0$  follows by definitional identity (`phi_md1_kink_bps_saturation := rfl`).

#### 9.4.1 EFT Operator Decomposition at $M_{\text{GUT}}$

At  $M_{\text{GUT}}$ , two physically distinct three-point operators contribute to the  $\text{top-}\Phi_{\text{MDL}}\text{-}N_1$  sector:

$$\mathcal{O}_1 = \bar{q}_L \tilde{H} t_R + \text{h.c.}, \quad Q_{\text{top}} = 0, \quad c_1 = y_t = 0.461 \quad (\mathbf{A}/\mathbf{D}); \quad (30)$$

$$\mathcal{O}_2 = \bar{t}_R \Phi_{\text{MDL}} N_R + \text{h.c.}, \quad Q_{\text{top}} = 1, \quad c_2 = g_{\text{kink-top}} = \varepsilon_{\text{FN}} \quad (\mathbf{A}_{\text{Lean}}). \quad (31)$$

$\mathcal{O}_1$  lies in the  $Q_{\text{top}} = 0$  (trivial topological) sector of the  $\Phi_{\text{MDL}}$  path integral and yields the SM top mass  $m_t = y_t v_{\text{EW}}/\sqrt{2} = 172.61 \text{ GeV}$  ( $\mathbf{A}/\mathbf{D}$ ).  $\mathcal{O}_2$  lies in the  $Q_{\text{top}} = 1$  BPS instanton sector and couples  $N_1$  to the top through the  $\Phi_{\text{MDL}}$  kink background. The two operators are distinguished by their field content ( $\tilde{H}$  vs.  $\Phi_{\text{MDL}}$ ) and by their  $\mathbb{Z}_7$  topological charge.  $\mathbb{Z}_7$  baryon-number conservation ( $\mathbf{A}_{\text{Lean}}$ ; `gte_baryon_number_topological_charge`) enforces  $\langle \mathcal{O}_1 \rangle_{Q_{\text{top}}=1} = 0$ , preventing mixing between the two sectors at tree level. The Wilson coefficient  $c_2 = g_{\text{kink-top}}$  therefore enters the Dirac Yukawa renormalization-group equation for  $N_1$  at  $M_{\text{GUT}}$  independently of  $y_t$ , giving  $c_2 = \varepsilon_{\text{FN}}$  from eq. (29) ( $\mathbf{A}/\mathbf{D}$ , combining  $\mathbf{A}_{\text{Lean}}$  inputs).

### 9.5 Kink excitation identification and leptogenesis

A consequence of identifying right-handed neutrinos as  $\Phi_{\text{MDL}}$  kink excitations: the  $\text{top-}\Phi_{\text{MDL}}$  coupling at the GUT scale equals the BPS instanton action,

$$g_{\text{kink-top}} = e^{-\pi/N_c} = \varepsilon_{\text{FN}} \quad (\mathbf{A}_{\text{Lean}}; \text{kink\_top\_coupling\_eq\_eps\_FN}), \quad (32)$$

where  $N_c = 3$  is the number of QCD colours and  $\varepsilon_{\text{FN}}$  is the Froggatt–Nielsen suppression factor derived from the  $Q = 1$  BPS instanton sector. This coupling enters the Dirac Yukawa renormalization-group equation at  $M_{\text{GUT}}$  in place of the SM top Yukawa  $y_t = 0.461$ ; the physical top mass  $m_t = 172.61 \text{ GeV}$  is unaffected, as it is governed by the top–Higgs coupling  $y_t$ . Combined with the leptogenesis calculation of P47 [8], this FKTT mechanism yields  $\eta_B = 6.109 \times 10^{-10}$  ( $+0.15\sigma$  from the Planck 2018 value). The formal derivation of eq. (32) is machine-certified ( $\mathbf{A}_{\text{Lean}}$ , zero sorry, zero axioms) in `FKTTCoupling.lean`:  $T_{11} = 0$  follows from the Bogomolny equation by definitional identity (`phi_md1_kink_bps_saturation := rfl`).

### 9.6 Open problem: C2-CLOSURE

**Conjecture 9.1** (C2, P34 [10]). *The physical coherence measure  $D \in [\mathbf{D}]$  is unique up to Lorentz and CPT symmetry.*

The CPT symmetry appearing in C2 has a precise algebraic characterisation: it is the unique non-trivial element of  $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}(\cos(2\pi/7))) \cong \mathbb{Z}_2$ , i.e., the Galois automorphism  $\sigma_6 : \zeta_7 \mapsto \zeta_7^6 = \bar{\zeta}_7$ . The decomposition  $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong \mathbb{Z}_2 \times \mathbb{Z}_3$  is  $\text{CPT} \times (\text{generation orbit symmetry})$  ( $\mathbf{A}/\mathbf{D}$ ; P44 [14], Lean: `cyclotomic_z7_master_bundle`, zero sorry). This provides a number-theoretic characterisation: CPT in GTE is a Galois automorphism of the kink moduli field, with  $h(-7) = 1$  (PID) and  $L(1, \chi_{-7}) = \pi/\sqrt{7}$  as arithmetic companions.

C2-CLOSURE would determine the specific value of  $\mathbb{P}_D^\top$  on non-trivial records, closing the remaining freedom in the two-function picture (§6.2). The present paper establishes that the *structure* of the selector (D3+D4+D5) is fixed without C2; what remains open is the specific  $D \in [\mathbf{D}]$ .

## 9.7 Holographic structure of domain walls

The  $\Phi_{\text{MDL}}$  domain walls in 3+1D realize a weak form of holographic bulk–boundary correspondence. A planar domain wall is a codimension-1 surface in 3+1D spacetime. Its energy scales as:

$$E_{\text{wall}} = \sigma \times \text{Area}, \quad (33)$$

so the wall entropy  $S_{\text{wall}} \propto \sigma \times \text{Area}$  — an area law consistent with the Bekenstein–Hawking formula  $S = A/(4l_P^2)$ . The  $\mathbb{Z}_7$  topological charge  $Q \in \mathbb{Z}_7$  is defined on the wall surface and constitutes the holographically encoded quantity: the bulk configuration in 3D space is determined (up to wall deformations) by the winding pattern on the 2D domain wall boundary.

This is the *algebraic* (weak) form of holography: the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, Theorem 8.1) is the precise mathematical realization of this correspondence — algebraic content on the 1D boundary ( $\mathbb{Z}_7$  topological charge sequence) encodes 3+1D physics. We do not claim the strong AdS/CFT state-level dictionary; the correspondence here operates at the level of the algebraic sector structure, consistent with the framework (**A**).

## 9.8 No-finite-CA uniqueness

The Lean-certified theorem `no_finite_ca_exact_lorentz_replica` (`CMCAContinuumLimit.lean`, **A**<sub>Lean</sub>, zero sorry) establishes that for all  $M > 0$ , the SR accuracy residual  $\varepsilon_0(M) = \pi^2/(3M^2) > 0$ . Combined with the existing `phimdl_lorentz_correction_tendsto_zero` (Theorem 8.4), this yields a strong uniqueness statement:

**Theorem 9.2** (Uniqueness of  $\Phi_{\text{MDL}}$ ; **A**<sub>Lean</sub>). *`phimdl_is_unique_exact_lorentz_model` in `CMCAContinuumLimit.lean`, zero sorry. Within the class of  $\mathbb{Z}_7$ -symmetric field models parameterized by lattice resolution  $M \geq 1$ , every finite- $M$  discrete model has strictly positive Lorentz invariance error. Only the  $M \rightarrow \infty$  continuum limit — which is  $\Phi_{\text{MDL}}$  — achieves exact Poincaré invariance.*

This result has a concrete physical interpretation: any lattice or discrete approximation to  $\Phi_{\text{MDL}}$ , however fine, will exhibit a residual Lorentz violation of order  $\pi^2/(3M^2)$ . The field  $\Phi_{\text{MDL}}$  itself is the unique model in this class with exact Poincaré symmetry.

## 9.9 QCA impossibility: winding-conserving coins must be diagonal

A linear quantum walk on  $\mathbb{Z}^3$  with  $\mathbb{Z}_7$  internal state — the closest discrete quantum analog of  $\Phi_{\text{MDL}}$  — avoids the vacuum/chaos bifurcation that plagues classical CAs in 3D, but cannot produce localized topological solitons. The Lean-certified theorem `commutes_with_winding_iff_diagonal` (`WindingCoinDecoupling.lean`, **A**<sub>Lean</sub>, zero sorry) establishes that any  $7 \times 7$  coin matrix  $C$  that commutes with the  $\mathbb{Z}_7$  winding operator  $W = \text{diag}(0, 1, \dots, 6)$  must be diagonal. The corollary `diagonal_coin_decouples_sectors` (**A**<sub>Lean</sub>) shows that diagonal coins decouple all winding sectors: the amplitude in sector  $s$  is scaled independently by  $e^{i\theta_s}$ , with no inter-sector quantum interference.

This is a structural impossibility result: topological localization requires nonlinear coupling between winding sectors, which is provided by the sine-Gordon potential  $V(\Phi) = (m_\phi^2/49)(1 - \cos 7\Phi)$



— the defining property of  $\Phi_{\text{MDL}}$ . Linear  $\mathbb{Z}_7$ -symmetric quantum walks cannot replicate the solitonic structure of  $\Phi_{\text{MDL}}$ , regardless of the choice of coin.

## 10 Quantum Corrections and Exact S-Matrix

### 10.1 One-loop effective potential and $\mathbb{Z}_7$ degeneracy

The Coleman–Weinberg effective potential at one loop preserves the  $\mathbb{Z}_7$  vacuum degeneracy:  $V_{\text{eff}}^{(1)}(\varphi_k) = -2.367 \times 10^{-2} \text{ GeV}^4$  at all seven  $\mathbb{Z}_7$  vacua (identical to within numerical precision; **A**, script `papers/42_phimdl_field/scripts/phimdl_cw_effective_potential.py`). The CW correction to the Higgs quartic  $\Delta\lambda_{\text{CW}} = -0.0175$  has the wrong sign to close the Higgs mass gap; the mechanism is ruled out as the  $\lambda$  correction (**A**).

### 10.2 Pöschl-Teller fluctuation spectrum

Small fluctuations  $\eta = \Phi - \Phi_{\text{kink}}$  about the BPS kink satisfy

$$V_{\text{fl}}(x) = m_\varphi^2 \left[ 1 - 2 \text{sech}^2(m_\varphi x) \right], \quad (34)$$

which is the Pöschl-Teller potential with  $s = 1$ . Machine-certified: `phimdl_fluctuation_is_poschl_teller` and `arctan_exp_cos_identity` (`PhiMDLFluctuationSpectrum.lean`, zero sorry, **A**<sub>Lean</sub>). The spectrum consists of a single zero mode and the continuum at  $k^2 > m_\varphi^2$ ; the potential is reflectionless. The integrals  $\int_{\mathbb{R}} \text{sech}^3 = \pi/2$  and  $\int_{\mathbb{R}} \text{sech} = \pi$  are Lean-certified from Mathlib FTC (`integral_sech_cubed`, `integral_sech`, zero sorry, **A**<sub>Lean</sub>).

*DHN formula inapplicable:* the  $\Phi_{\text{MDL}}$  coupling  $\alpha = 7 > \sqrt{8\pi} \approx 5.01$ , so the semi-classical DHN expansion does not apply. A 3+1D Casimir-type correction is required (see §10.4 below).

### 10.3 Exact S-matrix and TBA kernel

The kink–antikink scattering amplitude is identified with the  $\mathbb{ZZ}$  S-matrix in the sinh-Gordon limit  $B \rightarrow 1$ :

$$S(\theta) = \frac{\sinh \theta - i}{\sinh \theta + i}, \quad B = 1, \quad (35)$$

machine-certified (**A/D**; script `papers/42_phimdl_field/scripts/phimdl_casimir_tba.py`). The TBA (thermodynamic Bethe Ansatz) kernel follows from the S-matrix:

$$\varphi_{\text{TBA}}(\theta) = \frac{2}{\cosh \theta}, \quad (36)$$

(**A/D**; large- $u$  asymptotics  $u \cdot J - \pi/2 \sim -1/u$  verified).

### 10.4 Quantum kink mass

The quantum correction to the kink mass from the Pöschl-Teller spectrum in  $\overline{\text{MS}}$ -bar dimensional regularization employs the exact zero-mode coefficient  $C_{\text{zero}} = +1/3$  and the finite Casimir coefficient  $C_{\text{Cas}}$ . We denote the finite  $\overline{\text{MS}}$ -bar Casimir coefficient as

$$C_{\text{Cas}} = -0.707032607769(12) \quad (37)$$

This is the finite part of the dimensional regularization integral for the one-loop quantum mass correction of the  $s=1$  reflectionless Pöschl-Teller domain wall in 3+1 dimensions, renormalized in



the  $\overline{\text{MS}}$  scheme at scale  $\mu = m_\varphi$ . The value  $C_{\text{Cas}}$  does not reduce to any combination of standard mathematical constants ( $\pi$ ,  $\ln 2$ , Catalan’s constant, Euler–Mascheroni  $\gamma$ ,  $\zeta(3)$ , or algebraic multiples of  $1/\sqrt{2}$ ) to the precision computed; it constitutes a transcendental constant specific to the  $s=1$  reflectionless case. The quantum-corrected kink mass is

$$\Delta M = +31.22 \text{ MeV}, \quad M^Q = M_{\text{kink}} + m_\varphi \left( \frac{C_{\text{zero}}}{4\pi} + \frac{C_{\text{Cas}}}{8\pi^2} \right) = 321.32 \pm 15.6 \text{ MeV}, \quad (38)$$

(**A**;  $M^Q > M_{\text{kink}} = 290.10 \text{ MeV}$ ; scheme uncertainty  $\pm 15.6 \text{ MeV}$  from  $\mu \in [m_\varphi/2, 2m_\varphi]$ ). The prior log-UV cutoff gave  $\Delta M = -59.67 \text{ MeV}$  and  $M^Q = 230.43 \text{ MeV}$  (superseded). Scripts: `papers/42_phimdl_field/scripts/phimdl_casimir_3d1d_correction.py`, `phimdl_casimir_tba.py`.

## 11 Conclusion

We have established the quantum field theory of  $\Phi_{\text{MDL}}$  as the continuum companion to the Three-Layer Chiral Minkowski CA (P41). The principal results are:

1. **Poincaré invariance** (**A**<sub>Lean</sub>): the  $\Phi_{\text{MDL}}$  KG dispersion is Lorentz-invariant (machine-certified in `LorentzInvariance.lean`); the stress-energy tensor transforms as a rank-2 tensor (**A/D**, 400 random boosts verified).
2. **Born rule from field amplitudes** (**A/D/A**<sub>Lean</sub>):  $P(x) = |\partial_x \Phi|^2 / \int |\partial_x \Phi|^2$  normalizes exactly to 1; extended to overlapping kinks via the dual-frame inner product (`DualFrameBornRule.lean`); [**D**]-collapse implements projective measurement onto the kink eigenbasis.
3. **Physical quantum state** (**A**<sub>Lean</sub> conditional): the pre-measurement state is the Hamiltonian thermal ensemble on PSC-admissible sectors  $\{0, 2, 3, 4, 6\}$  (`PhiMDLThermalState.lean`); vacuum-dominant at observed cosmic temperatures; uniform at high  $T$ .
4. **3+1D extension** (**A/D/A**<sub>Lean</sub>): kinks become planar domain walls with tension  $\sigma = 290.10 \text{ MeV GeV}^{-2}$ ;  $\mathbb{Z}_7$  superselection and Born rule are unchanged; SR time dilation is exact. When two walls intersect, they form a junction line with exact tension  $\lambda_{\text{dim}} = -16/49$  (**A**<sub>Lean</sub>;  $\lambda = -1654.77 \text{ MeV fm}$ ; attractive,  $|\lambda/\sigma| = 2/m_\varphi$ ).
5. **Explicit descent map** (**A**): the Cook A-glider  $\mathbb{Z}_7$  winding profile matches the BPS kink with  $\text{RMSD} = 5.34\% < \varepsilon_0(7) = 6.71\%$  and Pearson  $r = 0.994$ , providing direct numerical confirmation of the  $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$  continuum limit.
6. **No-finite-CA uniqueness** (**A**<sub>Lean</sub>): every finite- $M$  discrete model has strictly positive Lorentz error;  $\Phi_{\text{MDL}}$  is the unique zero-error model in the  $\mathbb{Z}_7$ -symmetric class (`phimdl_is_unique_exact_lorentz_model`).
7. **QCA impossibility** (**A**<sub>Lean</sub>): any  $\mathbb{Z}_7$ -winding-conserving coin must be diagonal, decoupling all sectors and precluding quantum interference; topological localization is structurally impossible in a linear quantum walk (`commutes_with_winding_iff_diagonal`).
8. **Algebraic Lifting and Descent** (**A**<sub>Lean</sub>): the SM winding sector lifts to [**D**]-weighted states on  $\Phi_{\text{MDL}}$  (`algebraic_lifting_theorem`); all  $F_{21}$  conservation laws descend to any continuum realisation at every  $M \geq 1$  (`algebraic_descent_theorem`). The SR Nyquist correction  $\varepsilon_0(M) = \pi^2/(3M^2)$  is  $M$ -dependent and does not descend (`sr_accuracy_depends_on_M`).
9. **Continuum limit** (**A**<sub>Lean</sub> conditional):  $\varepsilon_0(M) = \pi^2/(3M^2) \rightarrow 0$  as  $M \rightarrow \infty$ ;  $\Phi_{\text{MDL}}$  is the unique terminal Level-1 GTE object (`CMCAContinuumLimit.lean`).
10. **Quantum corrections** (§10): Pöschl-Teller spectrum  $s = 1$  (**A**<sub>Lean</sub>; `phimdl_fluctuation_is_poschl_teller`);  $\mathbb{Z}_7$  one-loop degeneracy preserved (**A**); ZZ S-matrix  $S(\theta) = (\sinh \theta -$

$i)/(\sinh \theta + i)$  (**A/D**); TBA kernel  $\varphi(\theta) = 2/\cosh \theta$  (**A/D**); quantum kink mass  $M^Q = 321.32 \pm 15.6$  MeV (**A**, MS-bar dim-reg).

The two-function picture—Born rule  $\mathcal{B}$  and transputation selector  $\mathbb{P}^\top$ —provides a complete, axiom-minimal account of quantum measurement in the GTE framework, machine-certified in Lean 4 with zero sorry (conditional on the named **[D]** axioms of P34).

## A Lean 4 Theorem Inventory

All theorems listed below are in the `ugp-lean` repository (canonical public repo). Zero sorry. Disclosed axioms follow the theorem list.

**Table 3:** Lean 4 theorem inventory for P42. †: conditional on named axioms; see text.

| Theorem name                                        | Module                                            | S  | Content                                                    |
|-----------------------------------------------------|---------------------------------------------------|----|------------------------------------------------------------|
| <code>phimdl_kink_masses_equal</code>               | <code>Universality/PhiMDLThermalState</code>      | 0  | All $\mathbb{Z}_7$ kink masses equal                       |
| <code>psc_admissible_sectors</code>                 | <code>Universality/PhiMDLThermalState</code>      | 0  | $\{0, 2, 3, 4, 6\}$ admissible                             |
| <code>psc_admissible_card</code>                    | <code>Universality/PhiMDLThermalState</code>      | 0  | $\text{Card} = 5$                                          |
| <code>psc_sectors_partition</code>                  | <code>Universality/PhiMDLThermalState</code>      | 0  | $\text{Admissible} \cup \text{forbidden} = \mathbb{Z}_7$   |
| <code>thermal_state_sector_prob</code>              | <code>Universality/PhiMDLThermalState</code>      | 0  | $P(k \mid T) = e^{-M_k/T}/Z_T$                             |
| <code>sectorProb_admissible_sum</code>              | <code>Universality/PhiMDLThermalState</code>      | 0  | Normalization                                              |
| <code>vacuum_particle_prob_ratio</code>             | <code>Universality/PhiMDLThermalState</code>      | 0  | Vacuum dominance<br>$T \ll M_{\text{kink}}$                |
| <code>phimdl_thermal_state_master</code>            | <code>Universality/PhiMDLThermalState</code>      | 0  | Master bundle thermal state                                |
| <code>dual_gram_identity</code>                     | <code>Universality/DualFrameBornRule</code>       | 0  | $GG^{-1} = \mathbf{1}$                                     |
| <code>dual_frame_recovers_coefficients</code>       | <code>Universality/DualFrameBornRule</code>       | 0  | $a_k = c_k$                                                |
| <code>born_rule_from_dual_frame</code>              | <code>Universality/DualFrameBornRule</code>       | 0  | $\sum_k  a_k ^2 = 1$                                       |
| <code>born_rule_from_dual_frame_master</code>       | <code>Universality/DualFrameBornRule</code>       | 0  | Dual-frame Born master bundle                              |
| <code>transputation_is_minimizer</code>             | <code>Substrate/TransputationStateSelector</code> | 0  | $\mathbb{P}^\top = \arg \min D(\rho \mid w)$               |
| <code>transputation_minimizer_unique</code>         | <code>Substrate/TransputationStateSelector</code> | 0  | Uniqueness of minimum                                      |
| <code>two_function_picture</code>                   | <code>Substrate/TransputationStateSelector</code> | 0  | $\mathcal{B} \circ \mathbb{P}_D^\top$                      |
| <code>transputation_state_selector_bundle</code>    | <code>Substrate/TransputationStateSelector</code> | 0† | Two-function selector bundle                               |
| <code>poincare_invariance_of_kg</code>              | <code>Universality/LorentzInvariance</code>       | 0  | Full Poincaré                                              |
| <code>phimdl_lorentz_correction_tendsto_zero</code> | <code>Framework/CMCAContinuumLimit</code>         | 0  | $\varepsilon_0(M) \rightarrow 0$ as $M \rightarrow \infty$ |

(continued on next page)

(Table 3 continued)

| Theorem name                                | Module                                          | S              | Content                                                   |
|---------------------------------------------|-------------------------------------------------|----------------|-----------------------------------------------------------|
| cmca_continuum_limit_is_phimdl              | Framework/CMCAContinuumLimit                    | 0 <sup>†</sup> | CMCA $\rightarrow \Phi_{\text{MDL}}$ master               |
| born_rule_unconditional                     | Universality/<br>BeableWindingPartitionInstance | 0              | Sector Born rule                                          |
| algebraic_lifting_theorem                   | Spacetime/LiftingTheorem                        | 0              | PSC-admissible configs realised in $f_{\text{MDL}}$ (P28) |
| algebraic_descent_theorem                   | Universality/AlgebraicDescentTheorem            | 0              | $M$ -independence of $F_{21}$ content (P28)               |
| sr_accuracy_depends_on_M                    | Universality/AlgebraicDescentTheorem            | 0              | $\varepsilon_0(M)$ is $M$ -dependent (P28/P41)            |
| mkink_from_scc                              | Universality/GUTStructure                       | 0              | $M_{\text{kink}} = (8/49)m_\tau$ (P34/P35)                |
| phimdl_domain_wall_junction_tension_exact   | Substrate/WindingCoinDecoupling                 | 0              | $\lambda_{\text{dim}} = -16/49$ (exact)                   |
| phimdl_junction_is_attractive               | Substrate/WindingCoinDecoupling                 | 0              | $\lambda < 0$ : walls attract at junction                 |
| phimdl_junction_to_wall_ratio               | Substrate/WindingCoinDecoupling                 | 0              | $ \lambda/\sigma  = 2/m_\varphi$                          |
| commutes_with_winding_iff_diagonal          | Substrate/WindingCoinDecoupling                 | 0              | $\mathbb{Z}_7$ -winding-commuting coin iff diagonal       |
| diagonal_coin_decouples_sectors             | Substrate/WindingCoinDecoupling                 | 0              | Diagonal coin decouples all $\mathbb{Z}_7$ sectors        |
| no_finite_ca_exact_lorentz_replica          | Framework/CMCAContinuumLimit                    | 0              | $\varepsilon_0(M) > 0$ for all finite $M$                 |
| phimdl_is_unique_exact_lorentz_model        | Framework/CMCAContinuumLimit                    | 0              | $\Phi_{\text{MDL}}$ unique zero-error Lorentz model       |
| phimdl_axial_current_topological            | Substrate/ChiralCurrentL2                       | 0              | $\partial_\mu J_A^\mu = 0$ (Schwarz)                      |
| phimdl_vector_current_topological           | Substrate/ChiralCurrentL2                       | 0              | $\partial_\mu J_V^\mu = 0$ (Schwarz)                      |
| va_fraction_lifts_to_l2_chiral_current      | Substrate/ChiralCurrentL2                       | 0              | 32/125 lifts to L2 $J_A^\mu$                              |
| phimdl_l2_chiral_current_bundle             | Substrate/ChiralCurrentL2                       | 0              | G16 chiral-current bundle                                 |
| phimdl_weak_charged_current                 | Algebra/GaugeMDL                                | 0              | $J_W^\mu$ from $ D_\mu \Psi ^2$ (MDL gauging)             |
| weak_charged_current_bilinear_antisymmetric | Algebra/GaugeMDL                                | 0              | Isospin-exchange antisymmetry of $J_W^\mu$ bilinear       |
| phimdl_l2_chiral_and_weak_current_bundle    | Substrate/ChiralCurrentL2                       | 0              | G16+G18: V-A + weak charged current                       |

**Lean build verification.** All modules compile with:

```
cd ugp-lean
lake build UgpLean.Universality.PhiMDLThermalState
lake build UgpLean.Universality.DualFrameBornRule
```

```
lake build UgpLean.Substrate.TransputationStateSelector
lake build UgpLean.Framework.CMCAContinuumLimit
lake build UgpLean.Universality.LorentzInvariance
lake build UgpLean.Universality.BeableWindingPartitionInstance
lake build UgpLean.Spacetime.LiftingTheorem
lake build UgpLean.Universality.AlgebraicDescentTheorem
lake build UgpLean.Substrate.WindingCoinDecoupling
lake build UgpLean.Substrate.ChiralCurrentL2
lake build UgpLean.Algebra.GaugeMDL
```

Sorry count: zero (for all theorems in Table 3). Disclosed axioms: `d2_universal` (D2 of P34, PSC Presentation Invariance in universal form) in `PSCPILorentzMain.lean` and `CMCAContinuumLimit.lean`; `architectural_restriction` in `CoherenceMeasureUniqueness.lean`; the named physical-interpretation axioms of `GTECategoryStructure.lean`. The Born-rule modules `BeableWindingPartitionInstance.lean`, `DualFrameBornRule.lean`, and `PhiMDLThermalState.lean` carry zero custom physics axioms. The `WindingCoinDecoupling.lean` module contains two auxiliary `sorry`-carrying *lemmas* (`bps_kink_integral_eq_four` and `symmetry_integral_vanishes`) that are used only for internal documentation; all theorems cited in Table 3 are proved without these lemmas and carry zero `sorry`.

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